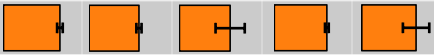
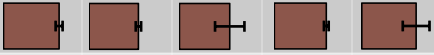
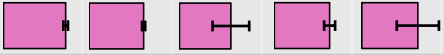
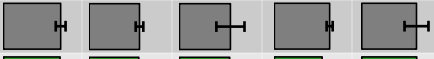
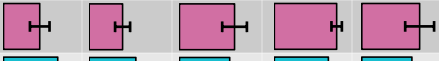
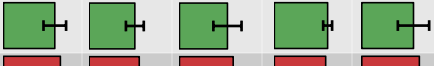
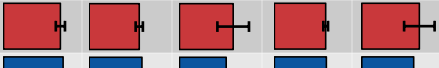
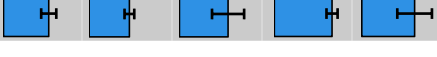
Methods	Core Algorithm/Model	How to estimate kinetic parameters		Whether to calculate and output du/dt	Performance at Cell level					Performance at Gene level				
					Local		Global			Gene-level kinetics			Kinetic parameters	
		α	β&γ		consistency score	ICCoh	CBDir	TransCosine	BATC	Real_CBDir	Sim_CBDir	Sim_CosSim	α_Spearman	γ_Spearman
scVelo_det	Steady-state linear regression	——	Gene-specific steady-state ratio (β/γ)	——						——	——	——	——	——
scVelo_sto	Stochastic model using second-order moments	——	Gene-specific moment fit	——						——	——	——	——	——
scVelo_dyn	Dynamical model with EM and analytical solution	Gene-specific	Gene-specific	Yes (model output)									——	——
MultiVelo (RNA_only)	RNA-only ODE with chromatin held fixed	Gene-specific	Gene-specific	Yes (model output)									——	——
UniTVelo	Unified-time dynamical model	Gene-specific RBF function	Gene-specific	Yes, not output									——	——
UniTVelo (mode2)	Independent-time dynamical model	Gene-specific RBF function	Gene-specific	Yes, not output						——	——	——	——	——
Dynamo	RNA velocity estimation + SparseVFC	Constant	β fixed γ gene-specific	——						——	——	——		——
TIvelo	Trajectory-guided velocity inference + orientation score	——	——	Explicit Output									——	——
cell2fate	Bayesian linearized ODE in Pyro	Module-weighted production	Gene-specific	——						——	——	——	——	——
TFvelo	Gene regulatory network + generalized EM	TF-regulated production	——	——						——	——	——	——	——
veloAE	Autoencoder representation learning	——	β fixed latent-space γ	——						——	——	——	——	——
veloVI	Deep generative model (VAE)	Latent state	Gene-specific	Yes (latent velocity output)									——	——
DeepKINET	Deep generative kinetic model	——	Cell-specific (VAE Output)	Yes (model output)						——	——	——	——	
TSvelo	Neural ordinary differential equation	Continuous function α(t)	Gene-specific	Yes (model output)						——	——	——		——
cellDancer	Deep neural network	Cell-specific	Cell-specific (DNN Output)	Yes (model output)										
DeepVelo	Graph convolutional network	Cell-specific	Cell-specific (GCN Output)	Yes (model output)										
SDEvelo	Stochastic differential equation	Sigmoid function	Gene-specific	——						——	——	——	——	——
DeepVelo_scv	Neural ODE + VAE reconstruction	——	——	from scVelo									——	——
GraphVelo_scv	Tangent-space projection + kNN velocity correction	——	——	from scVelo									——	——
GraphVelo_dyn	Tangent-space projection + kNN velocity correction	——	——	——						——	——	——	——	——